

PENGEMBANGAN SISTEM PERINGKASAN ULASAN OTOMATIS MENGUNAKAN TRANSFORMERS PADA APLIKASI TRAVELIO

S h a r o n Z e f a n y a S e t i a w a n / 2 5 0 1 9 6 1 0 2 2

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BAHASAN POKOK



✱ Pendahuluan

✱ Metode Penelitian

✱ Hasil dan Pembahasan

✱ Kesimpulan dan Saran



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PENDAHULUAN



LATAR BELAKANG

- Perkembangan teknologi meningkatkan jumlah aplikasi di pasaran.
- Pengguna menyampaikan kepuasan melalui ulasan di Google Play Store.
- Banyaknya ulasan membuat analisis menjadi sulit.
- Kesulitan ini menyulitkan pengguna dalam menentukan apakah aplikasi layak diunduh.

IDENTIFIKASI MASALAH

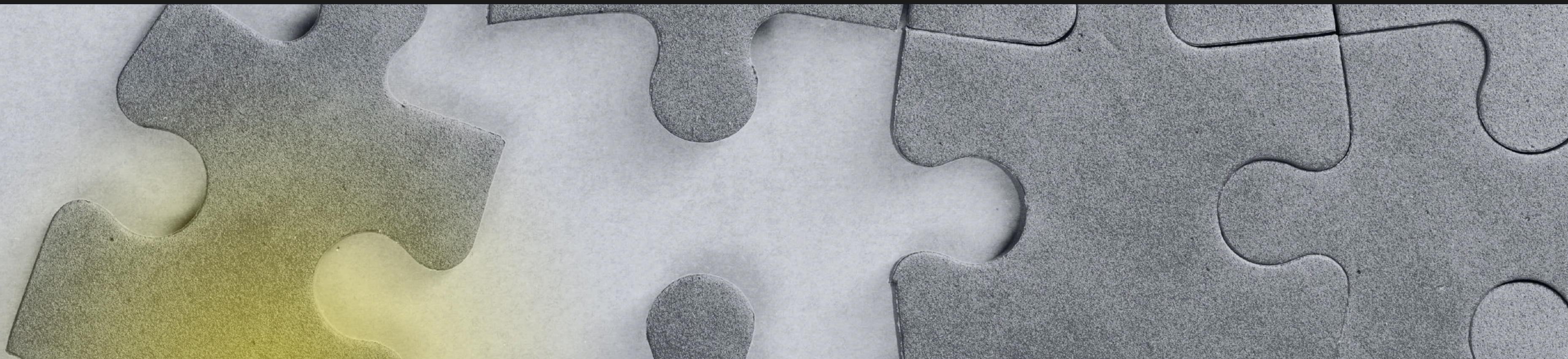
- ✱ Kebutuhan automasi peringkasan ulasan untuk menyediakan informasi relevan secara cepat dan efisien.
- ✱ Variasi struktur dan gaya penulisan ulasan menyulitkan pembuatan ringkasan yang koheren dan informatif.
- ✱ Kebutuhan meminimalisir bias dalam ringkasan agar memberikan wawasan yang seimbang.





SOLUSI

Mengembangkan sistem automasi peringkasan ulasan berbasis model Transformers dengan pendekatan hybrid text summarization dan memisahkan ulasan berdasarkan kategori sentimen (positif, netral, negatif) sehingga memberikan wawasan seimbang.



PENELITIAN PENDUKUNG



Paper	Materi	Deskripsi
[1]	<i>Long Document Summarization</i>	<i>Hybrid Model: BERT (Extractive) & BART (Abstractive)</i>
[2]	<i>Forum Discussion Summarization</i>	<i>Hybrid Model: T5</i>
[3]	<i>News Articles Summarization</i>	<i>Fine-Tuning BERT for Summarization</i>
[4]	<i>Indonesian News Articles Summarization</i>	<i>Abstractive method using BERTSUM</i>
[5]	<i>Food Reviews Summarization</i>	<i>Abstractive method using RNN</i>
[6]	<i>News Articles Summarization</i>	<i>Ensemble Learning using Pre-trained Models</i>

PENELITIAN PENDUKUNG



Identifikasi Kesenjangan:

- Belum ada penelitian yang berfokus untuk meringkas seluruh ulasan (*review*) dan menunjukkan seluruh aspek sentimen (positif, netral, negatif)
- Ketersediaan dataset ulasan untuk *text summarization* masih terbatas
- Dataset berbahasa Indonesia lebih sulit diproses

Penelitian yang Dilakukan:

- Fokus meringkas seluruh ulasan pengguna dan menampilkan seluruh aspek sentimen
- Dataset diambil menggunakan teknik *web scraping*
- Fokus pada dataset berbahasa Inggris

TARGET PENELITIAN



- Merangkum ulasan Travelio secara otomatis menggunakan model Transformers.
- Menangani variasi bahasa, gaya penulisan, dan panjang teks dalam komentar pengguna menggunakan model Transformers.
- Mengembangkan sistem peringkasan yang bermanfaat dan mudah diakses.



BATASAN MASALAH



- * Penelitian menggunakan Google Translate secara manual untuk standarisasi bahasa.
- * Penelitian tetap mengoreksi kesalahan terjemahan dan slang (bahasa gaul)
- * Model yang digunakan hanya fokus pada tugas peringkasan teks, tidak mencakup analisis sentimen.
- * Evaluasi model menggunakan metrik ROUGE, tanpa pertimbangan aspek kecepatan pemrosesan.
- * Sistem dikembangkan dengan Streamlit secara sederhana.

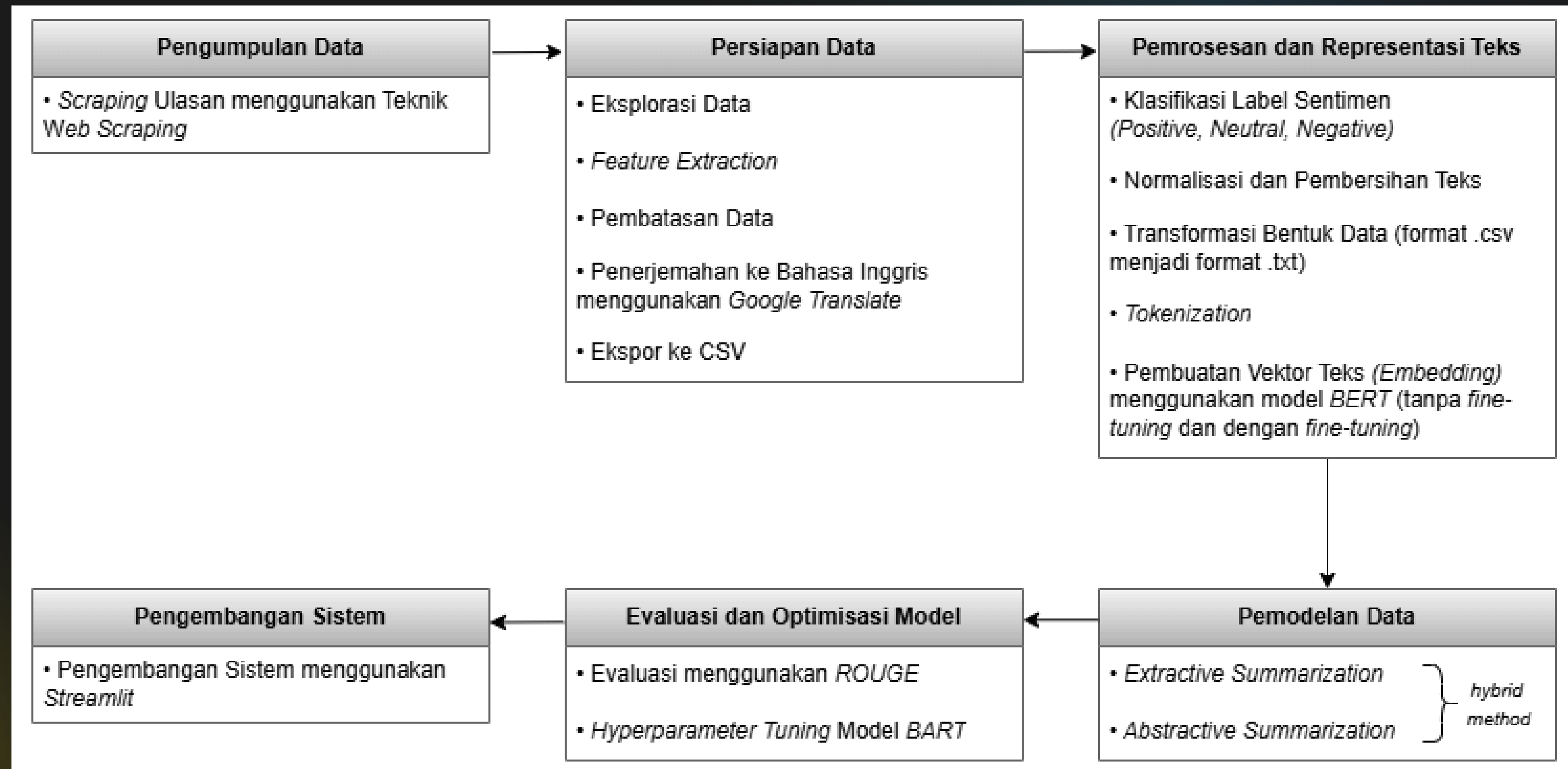


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METODE PENELITIAN

METODOLOGI PENELITIAN





DATA SET

- Ulasan aplikasi Travelio dalam bahasa Inggris.
 - Web scraping dari Google Play Store.
 - Jumlah Ulasan:
 - 2.044 ulasan positif
 - 106 ulasan netral
 - 1.403 ulasan negatif
- 
- 



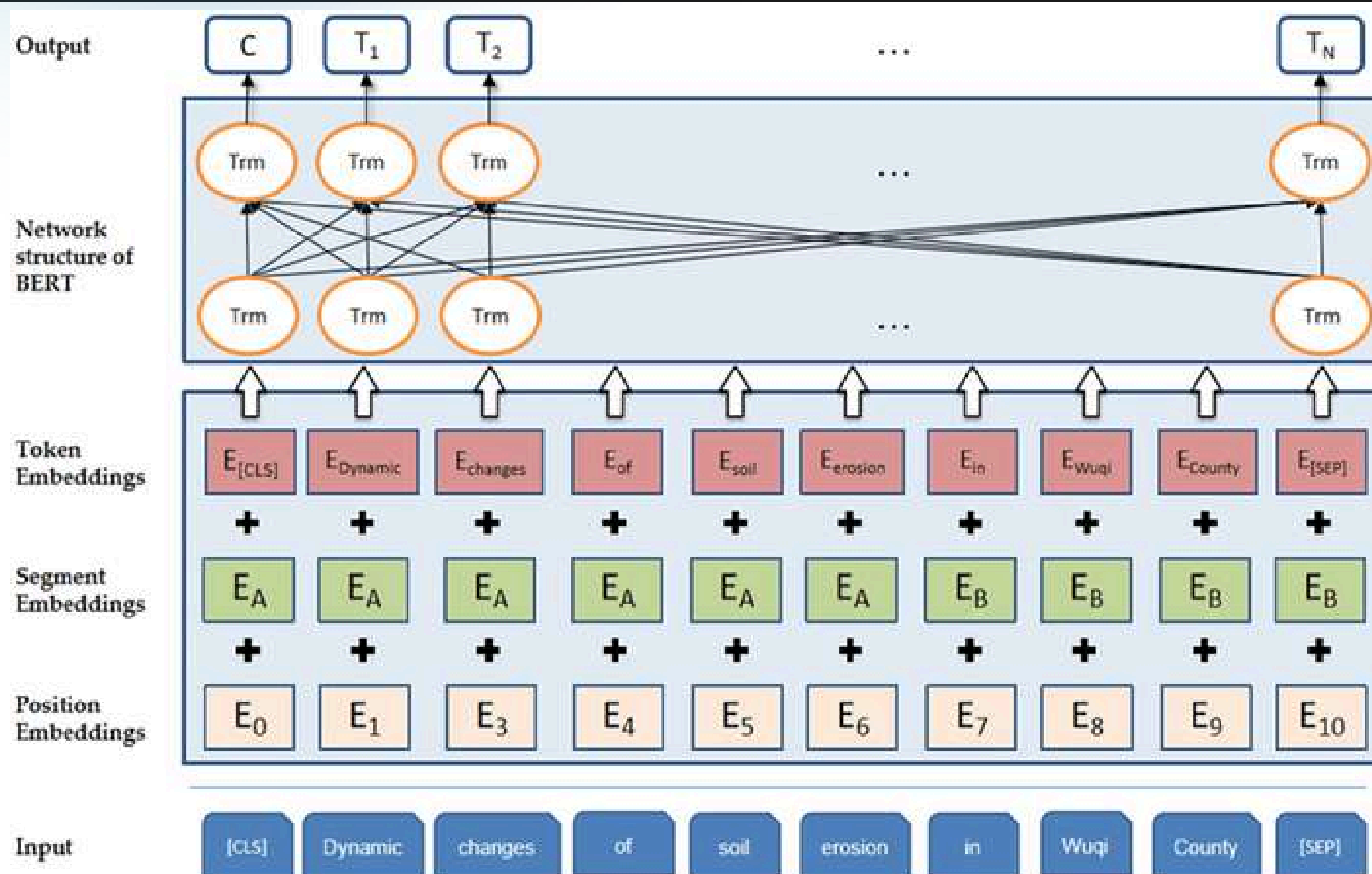
- 1 Konversi seluruh ulasan ke bentuk dokumen
- 2 Tokenisasi
- 3 Embedding menggunakan BERT + Fine-Tuning BERT
- 4 Peringkasan dengan metode ekstraktif menggunakan mean & max pooling
- 5 Hasil ekstraktif digenerasi kembali menghasilkan kalimat baru menggunakan BART (metode abstraktif)

ALUR HYBRID



MODEL BERT

(Bidirectional Encoder Representations from Transformers)



FINE-TUNING BERT



<i>Learning Rate</i>	2e-5
<i>Batch Size</i>	8
<i>Epoch</i>	3
<i>Optimizer</i>	<i>AdamW</i>
<i>Max Length</i>	512 token
<i>Early Stopping</i>	<i>True</i>



PENDEKATAN EKSTRAKTIF

top k = 5

Last Layer

Hidden states layer terakhir dari *BERT* adalah yang paling kontekstual.

Max Pooling

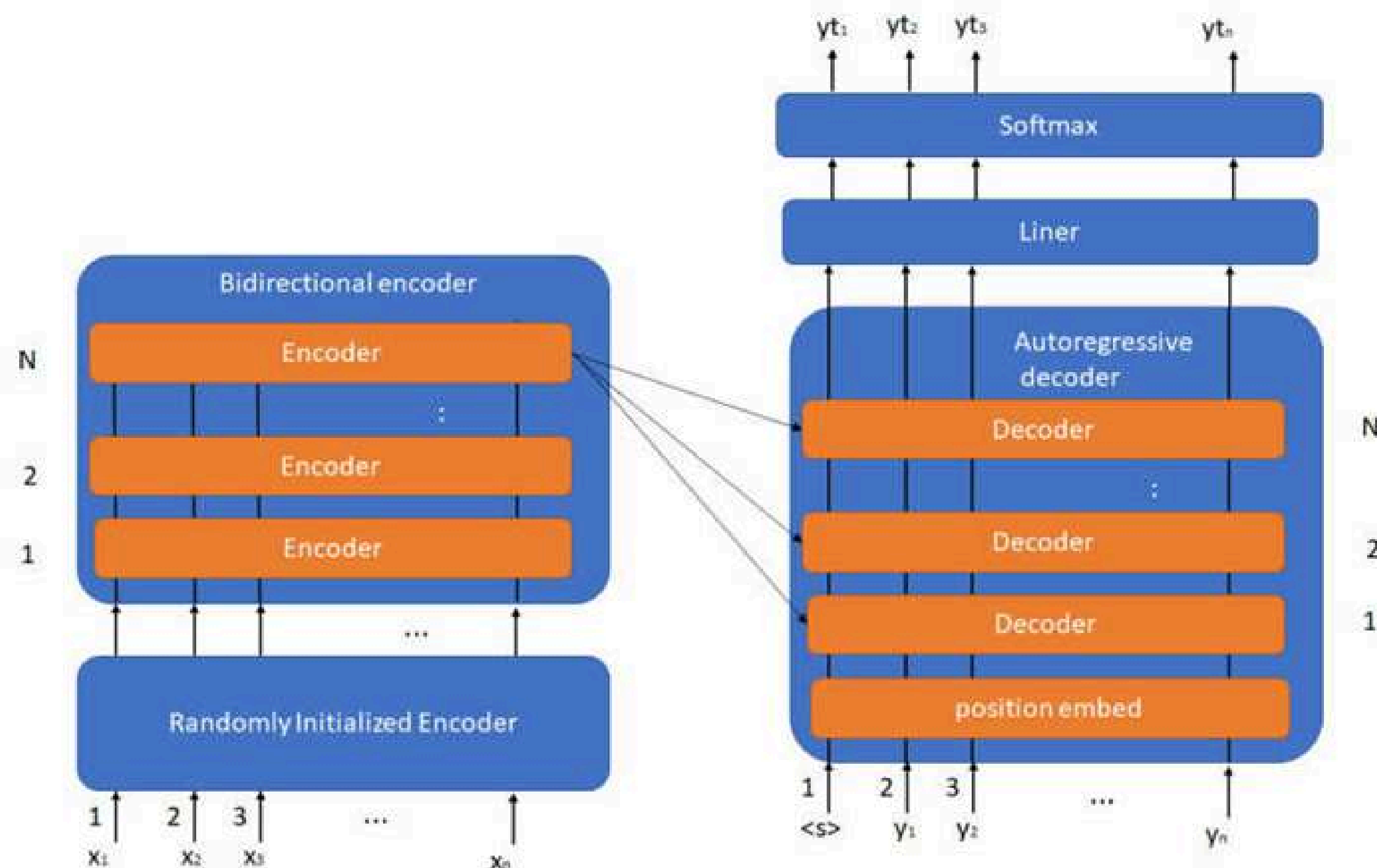
Setiap kalimat *embedding* dicari nilai tertinggiya untuk menemukan fitur yang paling dominan.

Mean Pooling

Kekuatan fitur dalam dataset diukur dengan menghitung rata-rata dari nilai maksimum tiap fitur.

MODEL BART

(Bidirectional and Auto Regressive Transformers)





METRIK EVALUASI

$$ROUGE_n = \frac{\sum_{S \in [\text{Reference summaries}]} \sum_{gram_n \in S} \text{Count}_{\text{match}}(gram_n)}{\sum_{S \in [\text{Reference summaries}]} \sum_{gram_n \in S} \text{Count}(gram_n)}$$

$$\text{Precision} = \frac{n(R \cap S)}{n(S)}$$

$$\text{Recall} = \frac{n(R \cap S)}{n(R)}$$

$$F1 - \text{Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Evaluasi menggunakan *F1-Score* terhadap referensi ringkasan dari satu Expert (Ahli Summarization) dan 30 responden umum.





BART

HYPERPARAMETER-TUNING

- Panjang Ringkasan:
 - Min Length: 20 token, lalu disesuaikan menjadi 50 token
 - Max Length: 50 token, lalu disesuaikan menjadi 150 token
- Panjang Penalti: 2.0
- Batas Pengulangan Frasa: 3
- Jumlah Pencarian Jalur: 4
- Early Stopping: True





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HASIL DAN PEMBAHASAN

FINE-TUNING BERT

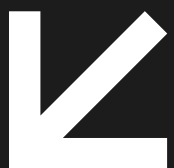
- Konvergensi baik dengan penurunan loss konsisten.
- Akurasi optimal meski ada variasi awal.
- BERT efektif mengenali sentimen.

	POSITIVE	NEUTRAL	NEGATIVE
Epoch 1/3			
Avg. Training Loss	0.0290	0.2985	0.0373
Validation Loss	0.0013	0.0995	0.0023
Validation Accuracy	1.00	1.00	1.00
Epoch 2/3			
Avg. Training Loss	0.0010	0.0692	0.0018
Validation Loss	0.0004	0.0364	0.0008
Validation Accuracy	1.00	1.00	1.00
Epoch 3/3			
Avg. Training Loss	0.0004	0.0287	0.0008
Validation Loss	0.0002	0.0163	0.0004
Validation Accuracy	1.00	1.00	1.00

EVALUASI MODEL



	Positive	Neutral	Negative
Without Fine-Tuning	The application makes it easy to find accommodation quickly and easily. There are many choices of units ranging from apartments villas houses. The customer service is responsive and quick to respond to any questions.	The service was okay from the start check in was a bit late but there was confirmation from the person for repairs complaints were served 1 x 24 hours air conditioner technicians technicians light repair cupboard rails each had their own expertise cleaning requests were	Travelio 700k is not worth it, says one user. "This is the first experience of using travelio and it s very disappointing from me"
Without Fine-Tuning and with Parameter Tuning	The application makes it easy to find accommodation quickly and easily. There are many choices of units ranging from apartments villas houses. The customer service is responsive and quick to respond to any questions or problems. There is also a discount promo for first bookings.	The service was okay from the start check in was a bit late but there was confirmation from the person for repairs complaints were served 1 x 24 hours air conditioner technicians technicians light repair cupboard rails each had their own expertise cleaning requests were easy and first cleaning times handled well if you have any questions they will be answered well via chat with us.	Travelio 700k is not worth it, says one user. "This is the first experience of using travelio and it s very disappointing from me," says another. "I prefer using airbnb the quality is more guaranteed really gone," says one.
With Fine-Tuning	The galeri ciumbleuit 3 apartment in bandung is a great place to stay. The staff were friendly and active. The linen smells great and the unit is clean.	The application is very good all the features for filtering prices locations and photos there are even 360 images but it is very unfortunate that my experience with the roomboy when checking in and checking out was very bad when i checked in i was charged	i stayed 4 nights in an apartment at the taman sari semanggi jakarta apartment i was told to deposit 200 thousand and promised to return it within 5 working days after checking out. 2 weeks have passed the deposit has
With Fine-Tuning and with Parameter Tuning	The galeri ciumbleuit 3 apartment in bandung is a great place to stay. The staff were friendly and active. The linen smells great and the unit is clean. There is wifi and you can cook too. The rental contract is clear.	The application is very good all the features for filtering prices locations and photos there are even 360 images but it is very unfortunate that my experience with the roomboy when checking in and checking out was very bad when i checked in i was charged a star on the spot he even asked to borrow my cellphone to make sure it had been reviewed when checking out the appointment was at 10 30 the day before then the next day until 12 00 they didn't show up finally self checkout. the air conditioner in the main room was broken the room boy was promised that at 9 o clock he could get service from a technician but it was already past 9 oclock when i called the call center.	i stayed 4 nights in an apartment at the taman sari semanggi jakarta apartment i was told to deposit 200 thousand and promised to return it within 5 working days after checking out. 2 weeks have passed the deposit has not been returned by travelio on the grounds that the last month s bill has not be issued. twice they canceled showings at the last minute after confirming earlier in the day and after i had taken time off work and traveled to the agreed location. give up on using this application the service is super bad.





ANALISIS MODEL

- Ulasan **Positif**: Model **tanpa fine-tuning** dengan parameter tuning → menghasilkan ringkasan universal.
- Ulasan **Netral**: Model dengan **fine-tuning tanpa parameter tuning** → menyeimbangkan pengalaman positif dan negatif.
- Ulasan **Negatif**: Model dengan **kombinasi fine-tuning dan parameter tuning** → menyajikan keluhan spesifik dan mendalam.



POSITIF

Hasil terbaik dicapai pada kondisi “**Without Fine-Tuning and with Parameter Tuning**” dengan Responden (Average) untuk ROUGE-1 sebesar 0.2665

NETRAL

Performa terbaik ditemukan pada kondisi “**With Fine-Tuning and without Parameter Tuning**” dengan Responden (Average) untuk ROUGE-1 sebesar 0.1868

NEGATIF

Performa optimal pada kondisi “**With Fine-Tuning and without Parameter Tuning**” untuk ROUGE-1 dari Expert dengan nilai 0.2222

			Positive	Neutral	Negative
Without Fine-Tuning and without Parameter Tuning	Expert	Rouge-1	0.2078	0.0964	0.1351
		Rouge-2	0.0000	0.0000	0.0000
		Rouge-L	0.1558	0.0482	0.0811
	Responden (Average)	Rouge-1	0.2682	0.1215	0.1705
		Rouge-2	0.0480	0.0119	0.0041
		Rouge-L	0.1873	0.0917	0.1187
Without Fine-Tuning and with Parameter Tuning	Expert	Rouge-1	0.2500	0.1346	0.1124
		Rouge-2	0.0000	0.0000	0.0000
		Rouge-L	0.1591	0.0577	0.0674
	Responden (Average)	Rouge-1	0.2665	0.1392	0.1592
		Rouge-2	0.0429	0.0092	0.0030
		Rouge-L	0.1734	0.0912	0.1141
With Fine-Tuning and without Parameter Tuning	Expert	Rouge-1	0.2254	0.1190	0.2222
		Rouge-2	0.0000	0.0488	0.0227
		Rouge-L	0.1408	0.0952	0.1111
	Responden (Average)	Rouge-1	0.2625	0.1868	0.1154
		Rouge-2	0.0219	0.0144	0.0022
		Rouge-L	0.1703	0.1277	0.0867
With Fine-Tuning and with Parameter Tuning	Expert	Rouge-1	0.2143	0.1065	0.2041
		Rouge-2	0.0000	0.0240	0.0138
		Rouge-L	0.1429	0.0710	0.0952
	Responden (Average)	Rouge-1	0.2367	0.1218	0.1425
		Rouge-2	0.0180	0.0084	0.0082
		Rouge-L	0.1598	0.0767	0.0906



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KESIMPULAN DAN SARAN



KESIMPULAN

- Model khusus per kategori lebih efektif dibandingkan model tunggal untuk semua kategori.
- Model Transformers (BERT & BART) mampu menangani variasi bahasa dengan mekanisme self-attention.
- Sistem peringkasan otomatis ulasan Travelio berhasil dikembangkan melalui tahapan sistematis.

SARAN

- Mencoba menangani ketidakseimbangan data.
- Mengeksplorasi model alternatif untuk meningkatkan kecepatan dan performa peringkasan.
- Mengembangkan model otomatis yang mengklasifikasikan sentimen dan menerjemahkan bahasa dalam satu *pipeline*.

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THANK YOU!

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